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Integrative analyses of the transcriptome and metabolome of Huzhang (*Pinellia pedatisecta* Schott) uncover the molecular genetic basis of its medicinal properties

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Abstract

Background *Pinellia pedatisecta* (Huzhang) is a traditional Chinese herb with a long medicinal history. Many metabolites with pharmacological activities are known to be present in the tissues of *P. pedatisecta*. However, the molecular basis of the biosynthesis of its pharmaceutical compounds is still unclear, which limits further studies on this valuable resource.

Results Based on untargeted metabolomic data, 640 metabolites were identified in leaves, stems, and tubers of *P. pedatisecta*, including 137 lipids, 107 organic acids and their derivatives, 88 phenylpropanoids and polyketides, and 7 alkaloids. Integrative analysis of the transcriptome and metabolome revealed several genes and metabolites related to the biosynthesis of amino acids, pyridine nucleotides, flavonoids, and polysaccharides. Additionally, some of these genes were found to be members of expanding gene families, confirming that the accumulation of corresponding metabolites in *P. pedatisecta* is regulated by the biosynthetic genes in this plant.

Conclusions Our present study provides valuable information on the genetic and medicinal properties of the *P. pedatisecta* herb, which will be significant for future utilization.

Keywords *Pinellia pedatisecta*, Metabolomic, Transcriptome, Amino acids, Pyridine nucleotides, Flavonoids, Polysaccharides

Background

Pinellia Tenore (Araceae) consists of nine perennial herbaceous species, mainly distributed in East Asia [1]. As an important distribution center, eight species of *Pinellia* have been recorded in China. Among these species, *P. ternata* (Chinese name: "Banxia") and *P. pedatisecta* (Chinese name: "Huzhang") are most widely used as traditional high-quality Chinese medicinal herbs since Eastern Han dynasty in the Divine Farmer's Materia Medica (Chinese name: "Shennong Bencao Jing") [2, 3]. Tubers produced by these plants have been traditionally used to treat vomiting, dispel wind and relieve convulsion, and eliminate stagnation [4]. Banxia (*P. ternata*) is listed in the Chinese Pharmacopoeia as a common traditional Chinese medicine.

Modern pharmacological studies have revealed that both *P. ternata* and *P. pedatisecta* possess a broad range of pharmacological properties. For example, *P. ternata*

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owns anti-cough, expectorant, and antiemetic effects [5, 6], and *P. pedatisecta* has antioxidant, anticonvulsant, anti-cancer, and anti-arrhythmic effects [7–9]. However, it has been reported that both species show proinflammatory toxicity, causing symptoms of sharp tingling sensation in the mouth and throat of people [10–12]. Owing to the morphological similarity of the dried tubers (the main medicinal tissues in *Pinellia* species), *P. ternata* and *P. pedatisecta* have been frequently used interchangeably in folk medicine, even if there are different medicinal properties and adverse effects [13].

The pharmacological effects of *P. ternata* and *P. pedatisecta* are closely related to the endogenous components, such as plant lectins, alkaloids, amino acids, nucleosides, and phytosterols [13]. However, the inadequate availability of these secondary metabolites has limited the development of quality standards for *P. ternata* and *P. pedatisecta*. Currently, more studies have focused on exploring the bioactive compounds of *P. ternata* [13, 14]. Also, several studies have characterized the biosynthetic pathways and related genes of ephedrine and acrid raphide formation, the key bioactive components of *P. ternata* [15, 16]. In the case of *P. pedatisecta* researches were mainly focused on the extraction, identification, and investigation of the pharmacological effects of a few active ingredients [17, 18]. Recently, Qian et al. [19] reported the genome of *P. pedatisecta* and fully characterized its lectin-related genes. However, compared with *P. ternata*, the information about the bioactive components of *P. pedatisecta*, particularly the molecular basis of the biosynthesis of the pharmaceutical compounds, is still insufficient; this vastly limits the use of this herb and further development of its products.

The untargeted metabolomics analysis aids in gathering a considerable amount of information on the metabolite composition of the biological samples [20]. This approach has been broadly used in the study of Chinese herbal medicines (CHMs) as it has significantly contributed to the comprehensive characterization of their metabolites and clarification of the complexity and integrity of CHMs [21–23]. With the development of omic technologies, the study of medicinal plants has been no longer limited to their chemical compositions, pharmacological properties, and pharmacodynamics; evaluating medicinal plants at the genetic level by integrating the multi-omic methods has become possible. Exemplary studies on *Pueraria lobata* var. *thomsonii* [24] and *Entada phaseoloides* [25] have validated that combined metabolomic and transcriptomic analyses can pinpoint key gene families and unravel molecular regulatory mechanisms of medicinal secondary metabolite biosynthesis. In addition, these omic resources help unravel the molecular basis of the medicinal plants to improve their pharmacological properties, further promoting the sustainable development of the Chinese medicine industry [26].

In this study, we initially explore the genes and metabolites associated with the biosynthesis of amino acids, pyridine nucleotide flavonoids, and polysaccharides based on a conjoint analysis of the transcriptome and metabolome of leaves, stems, and tubers of *P. pedatisecta.* This would provide new insight into the molecular breeding and medicinal properties of *P. pedatisecta*.

Materials and methods

Plant materials

The seeds of *P. pedatisecta* was collected at the Wuhan Botanical Garden, Chinese Academy of Sciences in Hubei Province, China and later planted in the plant greenhouse in March 2021 to subject the plants to the natural photoperiod. The plant material was formally identified by Prof. Jinming Chen. A voucher specimen has been deposited at the Herbarium of Wuhan Botanical Garden, CAS under the accession number HIB0258883. In May 2022, leaves, stems, and tubers were collected from *P. pedatisecta* with consistent growth including six biological replicates for metabolomic analyses (Fig. 1a). The samples were subsequently frozen in liquid nitrogen and stored at – 80 °C until use.

Metabolomic analyses

The untargeted metabolome analysis was carried out at Novogene Bioinformatics Technology Co. Ltd. (Beijing, China). In brief, leaves, stems, and tubers (~0.1 g) collected from six independent biological replicates were individually extracted with 500 µL of 80% pre-chilled methanol. The samples were incubated on ice for 5 min and then centrifuged at 15,000 × *g* at 4 °C for 15 min. An aliquot of the supernatant was diluted to a final concentration of 53% methanol with LC–MS-grade water. Further, the samples were centrifuged at 15,000 × *g* at 4 °C for 15 min. Finally, the supernatant was injected into the UHPLC-MS/MS system (ThermoFisher, Germany) for analysis. The raw data files generated by UHPLC-MS/MS were processed using the Compound Discoverer v3.1 (CD3.1, ThermoFisher) to perform peak alignment, peak picking, and quantitation for each metabolite. The main parameters were as follows: retention time tolerance, 0.2 min; actual mass tolerance, 5 ppm; signal intensity tolerance, 30%; signal/noise ratio, 3. After that, peak intensities were normalized to the total spectral intensity. The normalized data was used to predict the molecular formula based on additive ions, molecular ion peaks and fragment ions. And then, metabolite identification was performed by matching the peaks to the mzCloud (https://www.mzcloud.org/), mzVault, and MassList databases, achieving Level 2 confidence according to the Metabolomics Standards Initiative (MSI). Also, retain

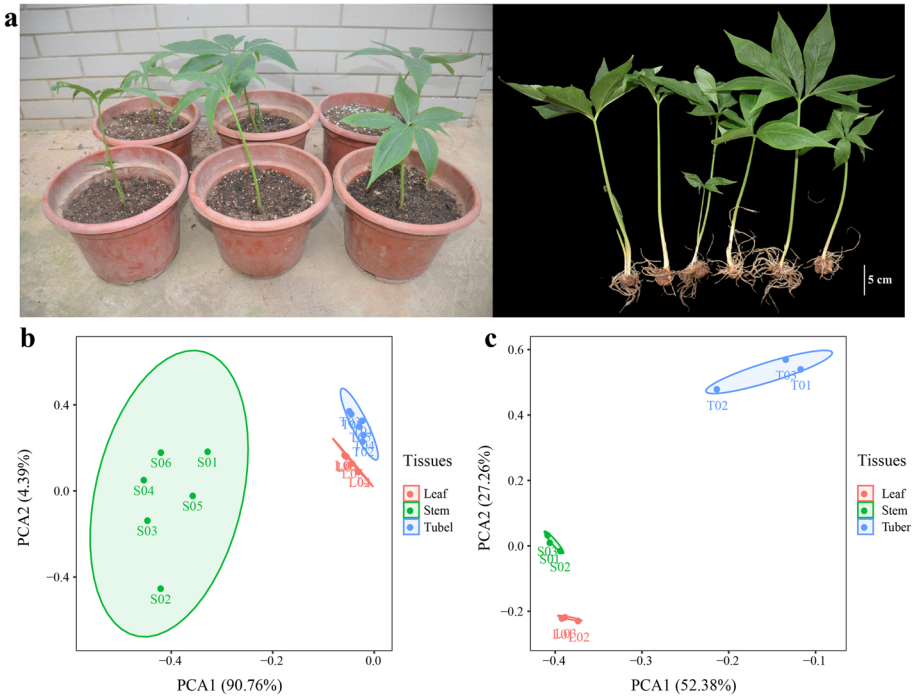


Fig. 1 Image of *P. pedatisecta* (**a**) and principal component analysis (PCA) of metabolite profiles (**b**) and transcriptome profiles (**c**) from different sample groups

metabolites with a coefficient of variance (CV) below 30% in the QC samples as the final identification results.

The metabolites with a value of $|\log_2$ (fold change) $|>1$ and a variable importance in projection (VIP) >1 were considered differentially abundant metabolites (DAMs). These metabolites were annotated using the KEGG (http://www.genome.jp/kegg/), HMDB (http://www.hmdb.ca/), and Lipidmaps databases (http://www.lipid maps.org/). The PCA and partial least squares discriminant analysis were performed using the program metaX [27]. The heatmap clustering of metabolite abundance values was conducted on the OmicShare online platform (http://www.omicshare.com/tools), with standardization via the Z-score method.

Transcriptomic data analysis

The transcriptome data of leaves, stems, and tubers from *P. pedatisecta* of our previous study was downloaded from the China National GeneBank DataBase (CNGBdb, https://db.cngb.org/) with accession number CNX0460292[19]. The metabolomic and transcriptomic data derived from the same biological samples. Briefly, clean RNA-seq data from nine samples of *P. pedatisecta* were aligned with the *P. pedatisecta* reference genome using HISAT2 v2.2.1 [28]. The gene expression levels were estimated using fragments per kilobase of exon per million fragments (FPKM) mapped using StringTie v1.3.3 [29]. DESeq2 [30] was used to analyze the differentially expressed genes (DEGs) in three comparisons, with cutoff values of $|\log_2$ (fold change) $|\geq 2$ and an adjusted *P*-value <0.01. In addition, the KEGG pathway enrichment analysis of DEGs were implemented using the OmicShare tool. To investigate the tissue-specific expression patterns of genes, we retained those with an average FPKM >1 across nine samples for further analysis. The tissue specificity was assessed using the previously proposed Tau index [31], and genes with Tau values ≥ 0.9 were considered to be tissue-specific.

Combined metabolomic and transcriptomic analysis

All the DEGs and DAMs obtained in Leaves_vs_Stems, Tubers_vs_Leaves, and Tubers_vs_Stems were mapped to the KEGG database to obtain their common pathway information. Furthermore, to determine the relationship between key genes and metabolites, the Pearson's correlation coefficient (PCC) was calculated between DEGs and DAMs using R v4.2.2 software. Only correlations with a $|\text{PCC}|\geq 0.9$ and an adjusted *P*-value for the Benjamini–Hochberg false discovery rate (FDR) <0.05 were selected.

Gene family analysis

The OG0000036 and OG0000333 gene family were obtained from the research by Qian et al. [19]. Then, these two gene families were multiple alignments by MUSCLE v3.8 [32], respectively. The NJ tree with 1000 bootstraps was constructed using protein alignments based on the JTT model in MEGA7 [33]. KaKs_Calculator v2.0 [34]

Table 1 Summary on metabolites of *P. pedatisecta* leaves, stems, and tubers

SuperClass	All	Leaf	Stem	Tuber
Alkaloids	7	3	6	5
Benzenoids	71	39	50	29
Lignans, neolignans and related compounds	7	1	5	4
Lipids	137	76	75	87
Nucleosides, nucleotides, and analogues	47	24	39	20
Organic acids and derivatives	107	63	82	56
Organic compounds	3	1	1	1
Organic nitrogen compounds	11	7	10	5
Organic oxygen compounds	51	25	35	25
Organoheterocyclic compounds	99	49	82	45
Organooxygen compounds	9	8	7	4
Phenylpropanoids and polyketides	88	54	58	36
Other	3	2	2	2
Total	640	352	452	319

was used to calculate non-synonymous substitutions (*Ka*) and synonymous substitutions (*Ks*) for members of the OG0000036 gene family.

Results and discussion

Metabolite profiles of leaves, stems, and tubers of P. pedatisecta and their KEGG pathway annotation

To understand the active medicinal components of *P. pedatisecta*, untargeted metabolomics were employed on the leaves, stems, and tubers. Principal component analysis (PCA) and hierarchical clustering analysis of the metabolite data obtained from all the tissues confirmed a high similarity of the metabolome from the same organ and revealed the tissue-specific accumulation patterns of the metabolites (Fig. 1b, S1). Moreover, 640 metabolites were detected in all the tissues, including 137 lipids, 107 organic acids and their derivatives, 88 phenylpropanoids and polyketides, and 7 alkaloids (Table S1). More metabolites (452) were found in the stems than in the leaves (352) or tubers (319; Table 1). Furthermore, 514 of these metabolites were annotated using the KEGG database. Most of them were enriched in the following metabolic pathways: amino acid metabolism, biosynthesis of other secondary metabolites, and nucleotide metabolism (Fig. S2).

Integrated analysis of transcriptome and metabolome

The clean RNA-seq reads were mapped to *P. pedatisecta* genome resulting in an average alignment rate of 96.55% (ranging from 93.75% to 97.36%, Table S2). The PCA results of the transcriptome data showed an acceptable variation among the biological replicates of each tissue (Fig. 1c), supporting experimental reproducibility. Based on the DEG and DAM analyses, a comparison between the leaf and the stem revealed 3132 DEGs, including 1215 upregulated and 1917 downregulated genes as well as 245 DAMs, including 32 upregulated and 213 downregulated metabolites (Fig. 2a, b). Moreover, a comparison between the tuber and the leaf revealed 5832 DEGs, including 3388 upregulated and 2444 downregulated genes as well as 173 DAMs, including 99 upregulated and 74 downregulated metabolites (Fig. 2c, d). Furthermore, a comparison between the tuber and the stem revealed 3844 DEGs, including 2401 upregulated and 1443

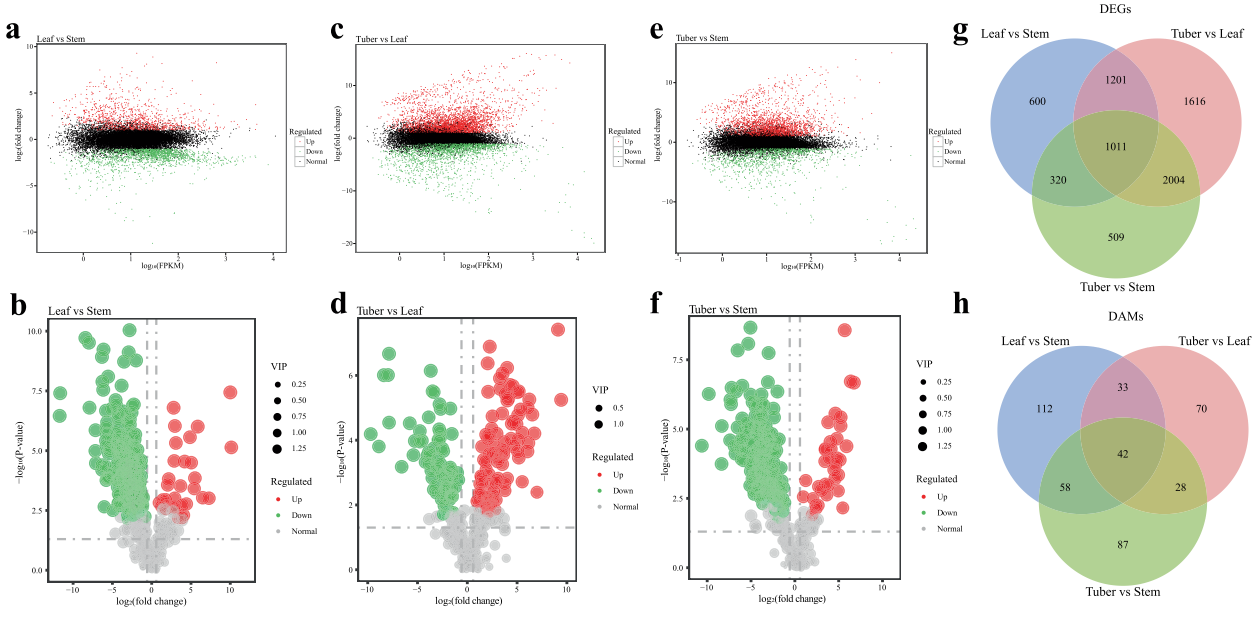


Fig. 2 Heatmap of metabolites and genes demonstrating the differences in their overall expression. **a** Differential abundant metabolites (DAMs) between the leaf and stem. **b** Differentially expressed genes (DEGs) between the leaf and stem. **c** DAMs between tuber and leaf. **d** DEGs between tuber and leaf. **e** DAMs between tuber and stem. **f** DEGs between tuber and stem. **g** Venn statistics of DEGs for all three pairs. **h** Venn statistics of DAMs for all three pairs

downregulated genes as well as 215 DAMs, including 35 upregulated and 180 downregulated metabolites (Fig. 2e, f). Simultaneously, 1011 DEGs and 42 DAMs were found to be common among the investigated tissues (Fig. 2g, h). The DEGs and DAMs identified from each comparative group were annotated using the KEGG database; we further determined the main biometabolic pathways in which DAMs and DEGs are commonly involved (Fig. S3). Notably, most KEGG pathways were closely related to amino acid biosynthesis and metabolism in the three comparative groups. The starch and sucrose metabolic pathway showed significant differences between the tuber and the leaf as well as between the tuber and the stem, which is potentially related to the tuber organ for starch storage. Also, previous comparative transcriptome analysis of various tissues of *P. ternata* revealed enrichment of genes associated with starch and sucrose metabolism in the tuber [16].

Identification of amino acids and the genes related to their biosynthesis

Previous studies have indicated that the main components responsible for the medicinal value of *P. pedatisecta* are their amino acids, nucleosides, and alkaloids [13, 35]. Amino acids are not only the components of proteins but are also precursors of many metabolites. Moreover, they have numerous functions in regulating the growth of organisms and other biological processes [36]. Currently, 15 amino acids have been identified in *P. pedatisecta* based on liquid chromatography-mass spectrometry [13]. Here, we have annotated 86 amino acids and their derivatives in *P. pedatisecta*, such as L-cystine, L-homocystine, methionine, and glutamine et al. (Fig. 3a). Based on the amino acid synthesis pathways reported in other species [37] as well as the transcriptomic and metabolomic data of *P. pedatisecta*, we proposed the amino acid biosynthesis pathway in *P. pedatisecta* for the first time. Amino acids are synthesized from glucose via the tricarboxylic acid (TCA) cycle, Phosphoribosyl diphosphate (PRPP) biosynthesis pathway, and pentose phosphate pathway (PPP). By aligning the genes of *P. pedatisecta* to the genes related to amino acid biosynthesis in *Arabidopsis thaliana*, we found 363 genes encoding 97 enzymes, of which 295 genes were expressed in the three types of tissues of *P. pedatisecta* (Table S3). Moreover, most of these enzymes were encoded by multiple genes. Of the 363 genes and 86 metabolites, 37 genes and one metabolite were associated with the TCA cycle, 16 genes and one metabolite were associated with the PPP, and five genes were associated with the PRPP biosynthesis pathway (Fig. 3b). Phenylalanine (Phe) biosynthesis is an important process in plant metabolism and physiology, due to its role as a building block for proteins and the precursor substance of the phenylethane metabolism [38]. Arogenate dehydratase (ADT) is the key enzyme responsible for phenylalanine biosynthesis, which contributes to the utilization rate of Phe [39, 40]. In this study, we found that *ADT* was highly expressed in the tuber (Fig. 3b), while Phe was also highest in tubers, indicating that the high expression of *ADT* as an upstream gene of Phe might promote the high accumulation of Phe in tubers.

To explore the regulatory network of genes on amino acid biosynthesis, Pearson's correlation analysis was performed between the expression levels of the candidate genes as well as the content of amino acids and their derivatives in the amino acid synthesis pathway. The results showed that nine genes were positively and significantly correlated with at least five amino acids (PCC > 0.9, FDR < 0.05), suggesting that these genes play key roles in regulating amino acid biosynthesis (Table S4).

Identification of pyridine nucleotides and the genes related to their biosynthesis

Nucleosides are important active compounds with irreplaceable antitumor and antiviral effects [41]. To date, seven nucleosides, including inosine, guanosine, adenosine, and thymidine have been isolated from *P. pedatisecta* [35]. In this study, 47 nucleosides were identified in *P. pedatisecta*. Among these, 39 nucleosides were detected in the stems, three nucleosides (5-methylcytidine, GDP-alpha-D-glucose, and ADP-ribose) were detected only in the tubers, and three nucleosides (dAMP, indole-3-carboxaldehyde, and cytarabine) were detected only in the leaves (Fig. 4a).

Pyridines are an important group of nucleotides. De novo synthesis of pyridine nucleotides begins with the conversion of aspartic acid to quinolinic acid. Nicotinamide adenine dinucleotide (NAD+), an important coenzyme in redox reactions, is involved in the de novo synthesis of pyridine nucleotides [42, 43]. Among the various types of NAD+ precursors, only β-nicotinamide mononucleotide shows improved pharmacokinetic and pharmacological properties, such as anti-aging and antioxidation effects [44]. In this study, these two metabolites were highly accumulated in the stems (Fig. 4a; Table S5). To further investigate their synthesis and metabolic pathways in *P. pedatisecta*, we analyzed the *P. pedatisecta* genes involved in the biosynthesis of pyridine nucleotides and the pyridine nucleotide cycle pathway. Moreover, the analysis revealed 12 genes encoding 12 enzymes (Fig. 4b). These results will be helpful for further development of the medicinal value of *P. pedatisecta.*

Identification of flavonoids and the genes related to their biosynthesis

Flavonoids are important phenolic compounds widely distributed throughout the plant kingdom. Flavonoids with distinct structures possess various important

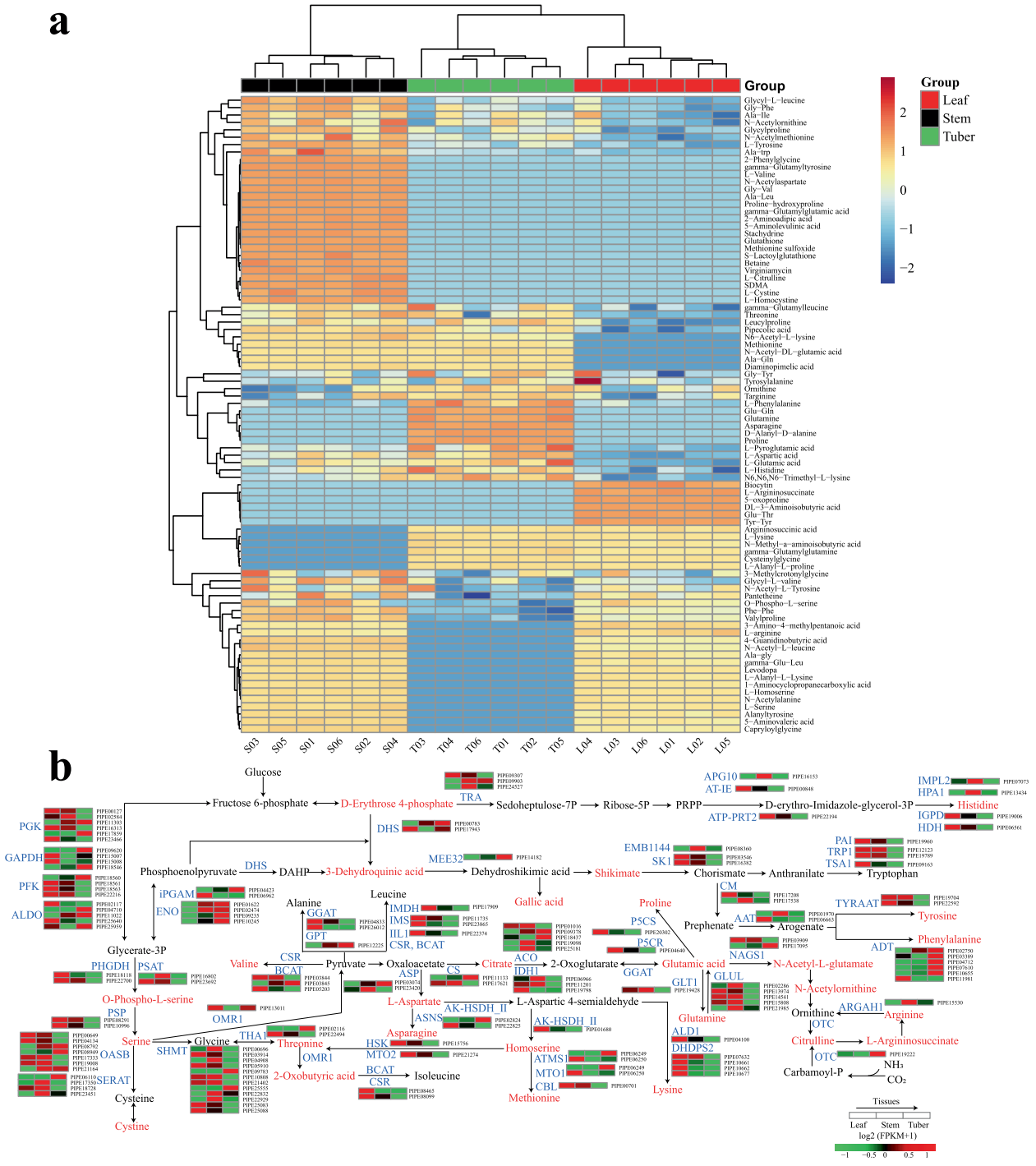


Fig. 3 The analysis of amino acid biosynthesis pathway of *P. pedatisecta*. **a** Heatmap of the amino acid metabolites identified in the three tissues. Each tissue had six biological replications. S (stem), T (tuber), L (leaf). **b** Schematic diagram of amino acid biosynthesis in *P. pedatisecta*. The red font indicates amino acid metabolites that have been identified in this study. The expression levels of the genes encoding enzymes for each sample are shown using the heatmap

pharmacological effects. For example, naringenin and naringin have anti-inflammatory and anti-cancer activities [45], and kaempferol shows anti-atherosclerotic and anti-allergic activities [46]. Previous studies have rarely focused on the flavonoids and related biosynthetic genes in *P. pedatisecta*. In this study, 34 flavonoids were identified in *P. pedatisecta*, including three 2-arylbenzofuran flavonoids, 24 flavonoids, and seven isoflavonoids (Table S1). We found that most flavonoids were identified in specific tissues; for instance, apigenin, isoliquiritin apioside,

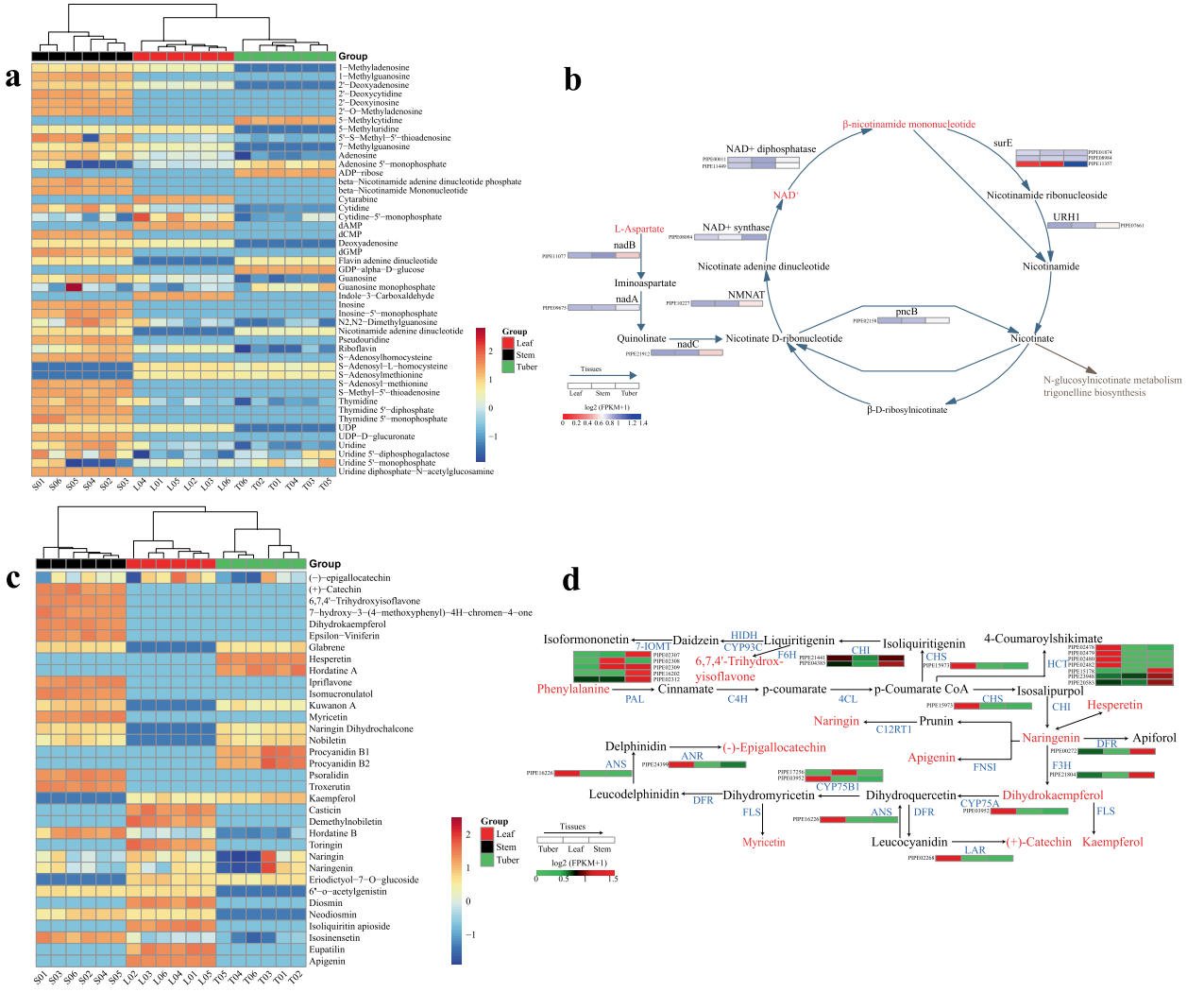


Fig. 4 The analysis of pyridine nucleotide and flavonoid biosynthesis pathway of *P. pedatisecta*. **a** Hierarchical cluster analysis of the pyridine nucleotide metabolites identified in the three tissues. Each tissue had six biological replications. S (stem), L (leaf), T (tuber). **b** Schematic diagram of pyridine nucleotide biosynthesis in *P. pedatisecta*. The expression levels of the genes encoding enzymes for each sample are shown using the heatmap. The red font indicates pyridine nucleotide metabolites that have been identified in this study. **c** Heatmap of the flavonoid metabolites identified in the three tissues. Each tissue had six biological replications. S (stem), L (leaf), T (tuber). **d** Schematic diagram of flavonoid biosynthesis in *P. pedatisecta*. The expression levels of the genes encoding enzymes for each sample are shown by the heatmap. The red font indicates flavonoid metabolites that have been identified in this study

and eriodictyol-7-O-glucoside were detected only in the leaves, and (+)-catechin, dihydrokaempferol, and ipriflavone were detected only in the stems. Moreover, kaempferol and eriodictyol-7-O-glucoside were detected in the leaves and tubers (Fig. 4c; Table S5). Hordatine B, isosinensetin, naringin, and naringenin were detected in all three tissues. It is noteworthy that previous studies identified flavonoids such as pelargonidin, vitexin, and prunin in *P. ternata* tubers [47], whereas these compounds were absent in *P. pedatisecta* in our study.

Further, 36 protein-coding genes involved in flavonoid biosynthesis were identified in *P. pedatisecta* using KEGG annotation (Table S6). Based on the identified genes and metabolites, we proposed a map of the biosynthesis pathways of the major flavonoid metabolites as well as the genes involved in these pathways in *P. pedatisecta* for the first time (Fig. 4d). Flavanone 3-hydroxylase (F3H) is a key enzyme of the flavonoid metabolic pathway that catalyzes the reaction of naringenin to form dihydroflavonol. Previous studies have indicated that high expression of *F3H* resulted in the accumulation of flavonoid metabolites in plant tissues [48–50]. In this study, we found that *F3H* was highly expressed in the stem and its downstream metabolites (dihydrokaempferol, catechin, myricetin, epigallocatechin) were similarly distributed in content, suggesting the accumulation of the above metabolites was closely related to the upstream transcription. In addition, seven of these genes (*PIPE02305*, *PIPE02306*, *PIPE02307*, *PIPE02308*, *PIPE02309*, *PIPE02312*, and *PIPE16202*) were found to belong to the OG0000036 gene family, which

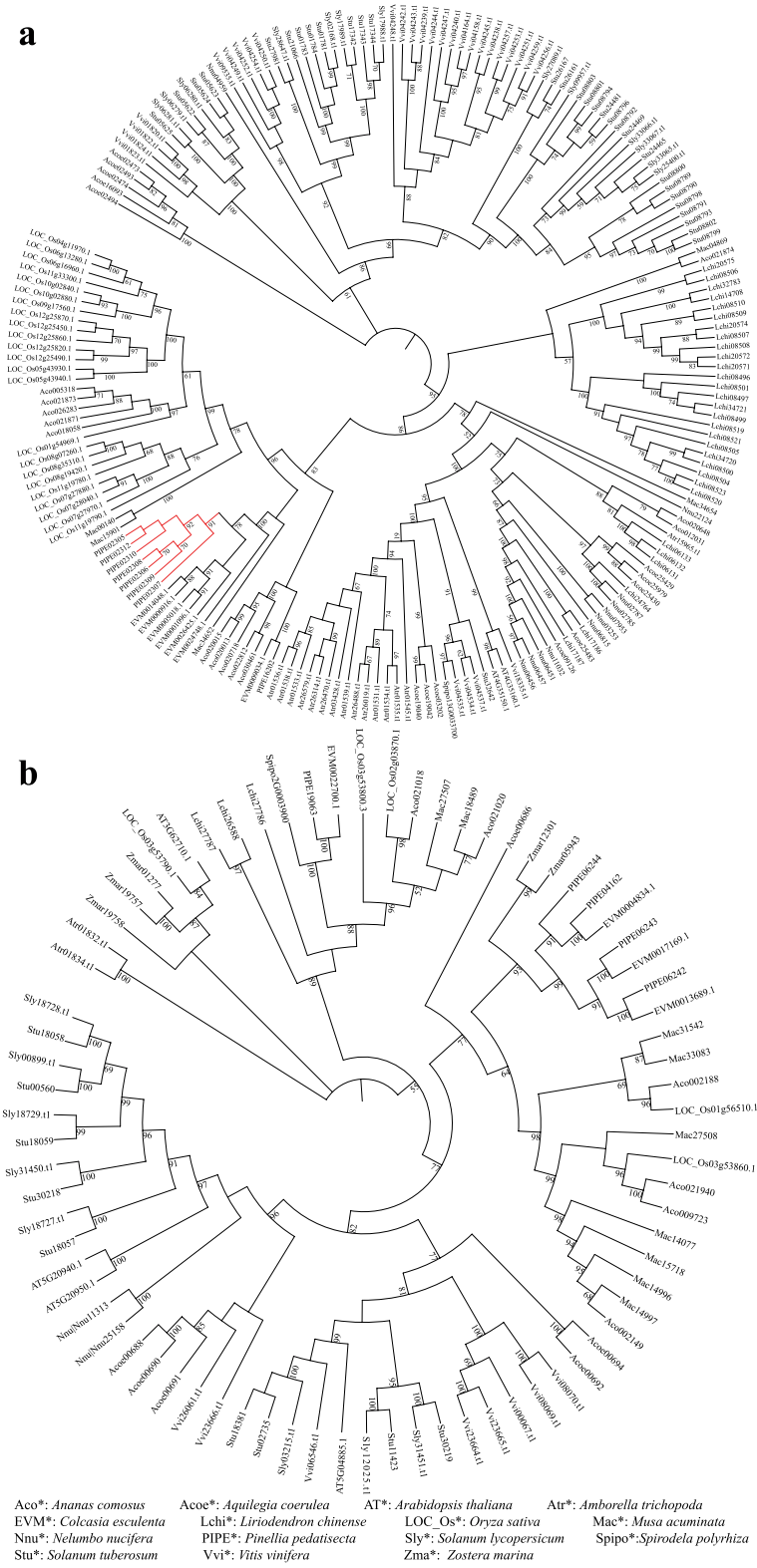


Fig. 5 Phylogenetic analysis of gene families. **a** OG0000036 gene family. Red branches denote lineage-specific expansion present in *P. pedatisecta*. **b** OG0000033 gene family. The NJ tree was constructed using protein alignments based on the JTT model in MEGA7. All bootstrap values (BS) that were higher than 50% are displayed

were predicted encodes isoflavone-7-O-methyltransferase [EC:2.1.1.150](Table S6). Five of the seven genes had tandem repeats. According to previous studies, the OG0000036 gene family has undergone expansion due to evolution [19]. Here, we constructed a phylogenetic tree for the OG0000036 gene family to reveal that species-specific expansions have occurred in *P. pedatisecta* after diverging from *Colocasia esculenta*, which may be associated with the recent tandem duplication events (Fig. 5a). Notably, the *Ka*/*Ks* ratios for all tandemly duplicated gene pairs were less than 1 (Table S7), indicating that most members of this gene family have underwent purifying selection to maintain their functionality. Furthermore, we found that two tandemly duplicated genes, *PIPE02307* (Tau = 0.98) and *PIPE02308* (Tau = 0.96), were specifically expressed in the stem and leaf, respectively (Fig. S4). The exact functions of these two genes in respective tissues remain to be further investigated.

Identification of polysaccharides and the genes related to their biosynthesis

Polysaccharides are one of the main components of herbal plants and have been proven to have important medicinal value [51]. In this study, we identified 45 polysaccharides in *P. pedatisecta*, most of which showed tissue-specific accumulation. Ribitol, D-glucose-6-phosphate, sucralose, and trehalose were highly accumulated in the leaves. Maltopentaose, maltotriose, GDP-alpha-D-glucose, D-sorbitol, ADP-ribose, and melibiose were highly accumulated in the tubers. In addition, isomaltose, D-glucarate, maltotetraose, D-glucosamine, and GDP-alpha-D-glucose were highly accumulated in the stems (Fig. S5; Table S5).

KEGG analysis here showed many protein-coding genes in *P. pedatisecta* that were involved in various pathways related to polysaccharide metabolism, including 209 genes associated with "starch and sucrose metabolism" and 71 genes associated with "fructose and mannose metabolism" (Table S8). Based on previous findings, five of these genes (*PIPE04162*, *PIPE06242*, *PIPE06243*, *PIPE06244*, and *PIPE19063*) were found to belong to the OG0000333 gene family, which putatively encodes β-glucosidase [EC:3.2.1.21] and has undergone expansion during evolution [19]. β-glucosidase is a member of the Glycosyl hydrolase family I, which can catalyze the hydrolysis of terminal non-reducing β-D-glucosyl residues to release β-D-glucose [52]. In this study, phylogenetic analysis of the OG0000333 gene family showed that these genes had high homology between *P. pedatisecta* and *C. esculenta* (Fig. 5b). *PIPE04162*, *PIPE06242* and *PIPE19063* were expressed in all tissues (Fig. S6). Moreover, many D-glucose derivatives were found in the three tissues of *P. pedatisecta*. These results provide a significant link between the genetic background of *P. pedatisecta* (e.g., expansion of gene families) and the accumulation of correlated metabolites in this plant.

Conclusion

Our study reported the high-throughput untargeted metabolomic data from leaves, stems, and tubers of *P. pedatisecta*. A total of 640 metabolites were identified across all tissues. We also systematically identified several metabolites with potential medicinal value, including 86 amino acids, 47 nucleosides, 34 flavonoids, and 45 polysaccharides. Additionally, the key candidate genes (e.g., *ADT* and *F3H*) associated with the biosynthesis of this medicinal plant have been identified. However, the specific functions of these genes still need to be verified. In summary, the metabolomic resources and candidate genes will be essential for further experimental verification and artificial synthesis of the active components of this medicinal plant in the future.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12864-026-12616-y.

Supplementary Material 1.

Supplementary Material 2.

Acknowledgements
Not applicable.

Authors' contributions
Z.H.Q. designed the study and led the research. C.P. and Z.H.Q. wrote the draft manuscript and analyzed the data. Z.H.Q. and J.M.C. contributed substantially to the revisions. The final manuscript has been read and approved by all authors.

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Data availability
The raw transcriptomic and metabolomic data that supports the findings of the current study have been deposited in the at the China National GeneBank DataBase (CNGBdb, https://db.cngb.org/) website under the accessions CNX0460292 and METM0000282.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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